Week 05 teacher guide

Decimals and long division

Keep this guide beside PowerPoint slides 255-315. Follow the numbered slide sequence, then use the matching answer section after the student has tried.

The seven days

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Decimal place value and products

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Weekly check

This week's idea

Decimal digits name parts of a whole. Addition and subtraction combine equal place-value units. Division preserves its quotient when both quantities scale equally.

Use the built-in Booster review where it appears in the day's work. It replaces ordinary practice. No separate Booster packet is needed.

PDF bookmarks jump to each lesson and answer section. The student workbook's Sunday reference pages are for checking an example after an attempt. Keep this teacher guide and the answer-review slides out of view during independent work.

Saturday. Decimal place value and products

Day 029, pages 116-120 | PowerPoint slides 255-272

Goal: I can add, subtract, and multiply decimals while keeping track of the value of each digit.

Set out: place-value paper, grid paper, pencil.

Check first: Whole-number multiplication, regrouping, and decimal place names.

Pacing: Retrieval 7; launch 8; explore 20; discuss 15; synthesis 7; exit 8; preview 5; slack 5. Total 75 min.

Explore: what to watch for

Aligns decimal points when adding. Why it matters: Like-sized units must be combined. Tutor move: Ask the student to name each column.

Estimates product size before computing. Why it matters: An estimate catches misplaced decimals. Tutor move: Ask whether taking a fraction of a fraction can exceed either factor.

Uses cents, fractions, or a grid. Why it matters: Several models correctly track decimal units. Tutor move: Accept any valid model and connect it to the written algorithm.

Likely misconceptions

Aligns the rightmost digits. Why: Treats every digit as the same unit. Tutor move: Write zeros to make place-value columns visible.

Makes a product larger than both small factors. Why: Keeps whole-number multiplication intuition. Tutor move: Take a part of one shaded part of a unit square.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 255. Week 5, Saturday

Topic: Decimal place value and products
Student: Watch and discuss

Say: I can add, subtract, and multiply decimals while keeping track of the value of each digit.

Ask: What do you already know about this idea?

Expect: Whole-number multiplication, regrouping, and decimal place names.

If stuck: Ask the student to name each column.

Slide 256. Today

Topic: Decimal place value and products
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

If stuck: Read the learning goal together.

Slide 257. Recall | 4 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 116 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 16; 2: 60 shells; 3: $\frac{8}{9}$ liter; 4: 21

If stuck: Give time to draw before giving a hint.

Slide 258. Recall: answers | 3 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 116 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 16; 60 shells; $\frac{8}{9}$ liter; 21

If stuck: Rebuild one missed item with its model.

Slide 259. Different decimal lengths | 8 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 117 · P1

Say: Read P1. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P1: 12.736

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 260. Subtract across zeros | 4 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 117 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P2: 7.315; P3: $9.40

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 261. The model | 4 min

Topic: Decimal place value and products
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: 0.6 × 0.4 = 24 hundredths.

If stuck: Ask the student to name each column.

Slide 262. Product below both factors | 3 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 118 · P4

Say: Read P4. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P4: 0.24; yes

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 263. Leading zeros in a product | 3 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 118 · P5, P6

Say: Read P5 and P6. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P5: 0.003; P6: $4.48

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 264. Watch the thinking | 3 min

Topic: Decimal place value and products
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 4.275

If stuck: Return to the matching unit in the model.

Slide 265. Your turn | 3 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 119 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: thousandths; 0.021

If stuck: Compare with the previous example without copying its numbers.

Slide 266. Two ways to see it | 8 min

Topic: Decimal place value and products
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Agree or disagree: Adding zeros at the end of 4.7 changes its value. Partner compare: How does a grid explain the decimal point in 0.6×0.4?

Expect: Accept drawings and equations that preserve the same quantities and units. Estimates product size before computing.

If stuck: Ask whether taking a fraction of a fraction can exceed either factor.

Slide 267. A claim to check | 7 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 119 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Six tenths times four tenths is twenty-four hundredths, or 0.24.

If stuck: Shade six tenths across and four tenths down on a hundred-square grid.

Slide 268. Words and meaning | 4 min

Topic: Decimal place value and products
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: The 4 in 0.04 means four hundredths. 6.2×3 is close to $6 \times 3 = 18$. 0.007 means seven thousandths.

If stuck: Read one definition and draw its example.

Slide 269. In your own words | 3 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 116 · S

Say: Use your own words and one example.

Ask: Agree or disagree: Adding zeros at the end of 4.7 changes its value.

Expect: Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

If stuck: Point to the model and name what each part represents.

Slide 270. Exit ticket | 8 min

Topic: Decimal place value and products
Student: Day 029 · Workbook p. 120 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 0.027

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 271. The week ahead | 5 min

Topic: Decimal place value and products
Student: Days 031-035 · Workbook pp. 131-140

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 272. Tutor reference

Topic: Decimal place value and products

Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 0.027

Got it: Gives 0.027 and supports its size with place value or a valid estimate.

Almost there: Gives 0.027 without a clear explanation, or uses correct units with an arithmetic slip.

Needs re-teaching: Gives 0.27, 2.7, or another value from counting places incorrectly.

Sunday. Long division with decimals

Day 030, pages 121-130 | PowerPoint slides 273-291

Goal: I can divide with decimals by preserving equal groups and checking my quotient.

Set out: place-value paper, pencil.

Check first: Whole-number multiplication and decimal place value.

Pacing: Retrieval 5; launch 5; explore 15; discuss 12; synthesis 5; mixed review 10; exit 8; preview 5. Total 65 min.

Explore: what to watch for

Scales both numbers equally. Why it matters: Preserves the number of groups. Tutor move: Ask for the factor written beside each number.

Adds zeros after a decimal point to continue. Why it matters: A remainder can be renamed in smaller units. Tutor move: Ask what three remaining ones become in tenths.

Checks quotient × divisor. Why it matters: The reverse operation tests correctness. Tutor move: Accept long division or equal-group reasoning with a multiplication check.

Likely misconceptions

Moves only one decimal point. Why: Uses a memorized instruction without equivalence. Tutor move: Rename both quantities in the same unit.

Stops at a nonzero remainder. Why: Does not yet rename the leftover amount. Tutor move: Write a decimal point and trade one unit for ten smaller units.

Follow the slide numbers below. Let the student attempt the task before showing the model or the answer. Write on the matching workbook pages; use extra paper only when more room is helpful.

Slide 273. Week 5, Sunday

Topic: Long division with decimals
Student: Watch and discuss

Say: I can divide with decimals by preserving equal groups and checking my quotient.

Ask: What do you already know about this idea?

Expect: Whole-number multiplication and decimal place value.

If stuck: Ask for the factor written beside each number.

Slide 274. Today

Topic: Long division with decimals
Student: Watch and discuss

Say: We will leave time to explain and check.

Ask: What can a model help us see?

Expect: Multiplying the dividend and divisor by the same power of ten preserves the quotient.

If stuck: Read the learning goal together.

Slide 275. Recall | 3 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 121 · R1, R2, R3, R4

Say: Work from memory. We will check together.

Ask: Which earlier model helps?

Expect: 1: 10 blue tiles; 2: 0, 12, and 21 centimeters; 3: 3; 4: 16

If stuck: Give time to draw before giving a hint.

Slide 276. Recall: answers | 2 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 121 · R1, R2, R3, R4

Say: Check one answer at a time.

Ask: Which step would you change?

Expect: 10 blue tiles; 0, 12, and 21 centimeters; 3; 16

If stuck: Rebuild one missed item with its model.

Slide 277. Three-digit by two-digit exact quotient | 5 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 122 · P1

Say: Try first. Then connect each quotient digit to place value.

Ask: How many groups of 24 fit in 67 tens?

Expect: P1: 28

If stuck: Estimate 672 ÷ 24 between 20 and 30. Use partial quotients 20 + 8 before recording the standard layout.

Slide 278. Whole numbers with decimal quotient | 3 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 122 · P2, P3

Say: Read P2 and P3. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P2: 3.375; P3: 1.2

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 279. The model | 3 min

Topic: Long division with decimals
Student: Watch and discuss

Say: Match each part of the model to a number in the problem.

Ask: What does one part represent?

Expect: Multiply both quantities by 100. The quotient stays 80.

If stuck: Ask for the factor written beside each number.

Slide 280. Decimal divided by decimal | 3 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 123 · P4, P5

Say: Read P4 and P5. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P4: 70; P5: 0.35

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 281. Divisor below one makes larger quotient | 2 min

Topic: Long division with decimals
Student: Day 030 · Workbook pp. 123, 124 · P6, P7

Say: Read P6 and P7. Give time to try before comparing methods.

Ask: What place does each digit name? What estimate should your answer be near?

Expect: P6: 9 pieces; P7: 42

If stuck: Write the numbers in matching units, or add zeros that keep the value unchanged.

Slide 282. Watch the thinking | 2 min

Topic: Long division with decimals
Student: Watch and discuss · worked example

Say: Read the reason beside each step.

Ask: Where does the model appear in the calculation?

Expect: 80

If stuck: Return to the matching unit in the model.

Slide 283. Your turn | 2 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 124 · F

Say: Complete the unfinished method.

Ask: Which step is still needed?

Expect: 540; 180

If stuck: Compare with the previous example without copying its numbers.

Slide 284. Two ways to see it | 6 min

Topic: Long division with decimals
Student: Watch and discuss · compare the same example

Say: Partner compare: show the same problem in two ways.

Ask: How does each part of the model match a step in the calculation? Agree or disagree: $3.6 \div 0.4$ must be less than 3.6 because it is division. Partner compare: Why do $4.8 \div 0.06$ and $480 \div 6$ count the same number of groups?

Expect: Accept drawings and equations that preserve the same quantities and units. Adds zeros after a decimal point to continue.

If stuck: Ask what three remaining ones become in tenths.

Slide 285. A claim to check | 6 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 124 · D

Say: Agree or disagree. Find the first step that fails.

Ask: Why does this mistake seem reasonable? How can a model check it?

Expect: Both amounts must scale by 100. Use 630 ÷ 9 = 70.

If stuck: Compare 63 tenths with nine hundredths by renaming both in hundredths.

Slide 286. Words and meaning | 3 min

Topic: Long division with decimals
Student: Watch and discuss

Say: Use each word while pointing to a model.

Ask: Which example shows the meaning most clearly?

Expect: 17 ÷ 5 gives 3 groups with 2 left. 6.4 ÷ 0.8 = 64 ÷ 8. 7 ÷ 2 = 3.5.

If stuck: Read one definition and draw its example.

Slide 287. Mixed review | 10 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 125 · M1, M2, M3

Say: Choose a method for each problem.

Ask: What tells you which operation to use?

Expect: $\frac{3}{8}$; 1.8000 dollars; $3\ \frac{1}{3}$

If stuck: Name the quantities and units before calculating.

Slide 288. In your own words | 2 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 121 · S

Say: Use your own words and one example.

Ask: Agree or disagree: 3.6 ÷ 0.4 must be less than 3.6 because it is division.

Expect: Multiplying the dividend and divisor by the same power of ten preserves the quotient.

If stuck: Point to the model and name what each part represents.

Slide 289. Exit ticket | 8 min

Topic: Long division with decimals
Student: Day 030 · Workbook p. 126 · E

Say: Work independently. Include the units and the requested reasoning.

Ask: Read the problem once. What is it asking you to find?

Expect: 120

If stuck: Do not coach during the ticket. Use the sort after collecting it.

Slide 290. The week ahead | 5 min

Topic: Long division with decimals
Student: Days 031-035 · Workbook pp. 131-140

Say: Each packet is a short practice session.

Ask: Which model might help you during the week?

Expect: A model connected to the current topics.

If stuck: Open the study guide to the matching example.

Slide 291. Tutor reference

Topic: Long division with decimals
Student: Tutor only · hidden in slide show

Tutor only. Keep this slide hidden during the lesson. Use it after collecting the exit ticket.

Exit answer: 120

Got it: Gives 120, writes 840 ÷ 7, and verifies 120 × 0.07 = 8.4.

Almost there: Uses the correct equivalent division but makes a minor calculation slip, or omits the check.

Needs re-teaching: Scales only one number, gives 1.2, or places the quotient decimal using a guess.

PowerPoint answer review

After Friday, outside the weekend lesson time. Say: Check one completed item at a time. Ask: Where does your method first differ from this one? Expect: The answers and worked reasoning printed in the complete keys below. If stuck: Rebuild the quantity with its model, then have the student repair the first incorrect step.

Keep these slides out of view during independent work. The index below follows the answer-review slides in order; every answer is also printed in the complete keys.

Slide 292. Week 5 answers

Next lesson: slide 316. Use this review section after Friday.

Slide 293. Decimal place value and products (1/4)

Items: S09:R1; S09:R2; S09:R3; S09:R4. Follow along: Day 029 · Workbook p. 116 · Items R1, R2, R3, R4

Slide 294. Decimal place value and products (2/4)

Items: S09:S; S09:P1; S09:P2; S09:P3. Follow along: Day 029 · Workbook pp. 116, 117 · Items S, P1, P2, P3

Slide 295. Decimal place value and products (3/4)

Items: S09:P4; S09:P5; S09:P6; S09:F; S09:D. Follow along: Day 029 · Workbook pp. 118, 119 · Items P4, P5, P6, F, D

Slide 296. Decimal place value and products (4/4)

Items: S09:E. Follow along: Day 029 · Workbook p. 120 · Items E

Slide 297. Long division with decimals (1/4)

Items: S10:R1; S10:R2; S10:R3; S10:R4. Follow along: Day 030 · Workbook p. 121 · Items R1, R2, R3, R4

Slide 298. Long division with decimals (2/4)

Items: S10:S; S10:P1; S10:P2; S10:P3. Follow along: Day 030 · Workbook pp. 121, 122 · Items S, P1, P2, P3

Slide 299. Long division with decimals (3/4)

Items: S10:P4; S10:P5; S10:P6; S10:P7; S10:F. Follow along: Day 030 · Workbook pp. 123, 124 · Items P4, P5, P6, P7, F

Slide 300. Long division with decimals (4/4)

Items: S10:D; S10:M1; S10:M2; S10:M3; S10:E. Follow along: Day 030 · Workbook pp. 124, 125, 126 · Items D, M1, M2, M3, E

Slide 301. Study guide: Try it

Items: W05_Study_Try:1; W05_Study_Try:2; W05_Study_Try:3; W05_Study_Try:4; W05_Study_Try:5. Follow along: Day 030 · Workbook p. 130 · Items 1, 2, 3, 4, 5

Slide 302. Daily practice (1/3)

Items: Day031_Practice:1; Day031_Practice:2; Day031_Practice:3; Day031_Practice:4; Day031_Practice:5. Follow along: Day 031 · Workbook p. 131 · Items 1, 2, 3, 4, 5

Slide 303. Daily practice (2/3)

Items: Day031_Practice:6; Day031_Practice:7; Day031_Practice:8; Day031_Practice:9; Day031_Practice:10. Follow along: Day 031 · Workbook pp. 131, 132 · Items 6, 7, 8, 9, 10

Slide 304. Daily practice (3/3)

Items: Day031_Practice:11; Day031_Practice:12; Day031_Practice:13. Follow along: Day 031 · Workbook p. 132 · Items 11, 12, 13

Slide 305. Daily practice (1/3)

Items: Day032_Practice:1; Day032_Practice:2; Day032_Practice:3; Day032_Practice:4; Day032_Practice:5. Follow along: Day 032 · Workbook p. 133 · Items 1, 2, 3, 4, 5

Slide 306. Daily practice (2/3)

Items: Day032_Practice:6; Day032_Practice:7; Day032_Practice:8; Day032_Practice:9; Day032_Practice:10. Follow along: Day 032 · Workbook pp. 133, 134 · Items 6, 7, 8, 9, 10

Slide 307. Daily practice (3/3)

Items: Day032_Practice:11; Day032_Practice:12; Day032_Practice:13. Follow along: Day 032 · Workbook p. 134 · Items 11, 12, 13

Slide 308. Daily practice (1/3)

Items: Day033_Practice:1; Day033_Practice:2; Day033_Practice:3; Day033_Practice:4; Day033_Practice:5. Follow along: Day 033 · Workbook p. 135 · Items 1, 2, 3, 4, 5

Slide 309. Daily practice (2/3)

Items: Day033_Practice:6; Day033_Practice:7; Day033_Practice:8; Day033_Practice:9; Day033_Practice:10. Follow along: Day 033 · Workbook pp. 135, 136 · Items 6, 7, 8, 9, 10

Slide 310. Daily practice (3/3)

Items: Day033_Practice:11; Day033_Practice:12; Day033_Practice:13. Follow along: Day 033 · Workbook p. 136 · Items 11, 12, 13

Slide 311. Daily practice (1/3)

Items: Day034_Practice:1; Day034_Practice:2; Day034_Practice:3; Day034_Practice:4; Day034_Practice:5. Follow along: Day 034 · Workbook p. 137 · Items 1, 2, 3, 4, 5

Slide 312. Daily practice (2/3)

Items: Day034_Practice:6; Day034_Practice:7; Day034_Practice:8; Day034_Practice:9; Day034_Practice:10. Follow along: Day 034 · Workbook pp. 137, 138 · Items 6, 7, 8, 9, 10

Slide 313. Daily practice (3/3)

Items: Day034_Practice:11; Day034_Practice:12; Day034_Practice:13; Day034_Practice:14. Follow along: Day 034 · Workbook p. 138 · Items 11, 12, 13, 14

Slide 314. Weekly check (1/2)

Items: Day035_Practice:1; Day035_Practice:2; Day035_Practice:3; Day035_Practice:4; Day035_Practice:5. Follow along: Day 035 · Workbook p. 139 · Items 1, 2, 3, 4, 5

Slide 315. Weekly check (2/2)

Items: Day035_Practice:6; Day035_Practice:7; Day035_Practice:8; Day035_Practice:9; Day035_Practice:10. Follow along: Day 035 · Workbook p. 140 · Items 6, 7, 8, 9, 10

Weekday teaching and review

Use one day at a time. Aim for 15-20 minutes, or 15 minutes during a holiday week. Accuracy and a clear explanation matter more than finishing quickly.

Day 031. Monday | pages 131-132

Ask for a model from Saturday before checking an answer. Let the student choose a method. After checking, have the student correct the first step that changed the quantity.

Built-in Booster review: Decimal practice: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 032. Tuesday | pages 133-134

Connect the practice to Sunday's model. Ask what each number counts and how a multiplication check, diagram, or estimate supports the answer.

Built-in Booster review: Multiplication and long division: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 033. Wednesday | pages 135-136

Let the student choose the method for each mixed problem. For the puzzle, ask which conditions are already met. Accept any solution that meets every condition.

Built-in Booster review: Weekly puzzle: items 13. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 034. Thursday | pages 137-138

Ask the student to explain the planted error before repairing it. Optional Stretch comes after the required work and may be skipped without delaying the next day.

Built-in Booster review: Fraction operations: items 9, 10, 11, 12. These items replace ordinary practice; they are not extra homework.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

Day 035. Friday | pages 139-140

Give the weekly check independently, without coaching or a calculator. Score the required reasoning as well as the answer. Afterward, choose one error to discuss calmly.

If the same skill causes three misses, stop and revisit its model. Use the included review item as a teaching example, then try it again later without the explanation. Do not add a separate worksheet.

After Friday

On a 10-item check: 9-10 correct is strong; 7-8 is on track; 5-6 needs another review of the missed topic; 0-4 calls for 20 minutes of re-teaching at the next session. Record the skill and one observation. Check it again with a fresh item at least a week later.

Complete answers for the week

Match the Day number, item ID, and student page before checking. Questions stay in the student workbook; these keys give every answer and the needed reasoning. Equivalent correct forms and sound alternative methods are welcome. An explanation item needs a reason, model, or valid check as well as its conclusion.

Decimal place value and products

Day 029, student workbook pages 116-120

R1 [R-TABLE] | student p. 116

Answer: 16

Reasoning: Multiply both entries in the first pair by the same factor.

R2 [P-WHOLE] | student p. 116

Answer: 60 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 5.40 shells: Taking 30% of the known part does not recover the whole.

R3 [F-DIV] | student p. 116

Answer: $\frac{8}{9}$ liter

Reasoning: One fourth holds two ninths liter. Four fourths hold eight ninths liter.

R4 [R-TABLE] | student p. 116

Answer: 21

Reasoning: Multiply both entries of the first pair by three.

S [EXPLAIN] | student p. 116

Answer: Answers vary. Addition combines like-sized units. Multiplication combines both the number of parts and the size of each part.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [D-ADD] | student p. 117

Answer: 12.736

Reasoning: Write 4.700 so tenths, hundredths, and thousandths line up.

Watch for 8.083: Aligning the final digits combines different units.

P2 [D-SUB] | student p. 117

Answer: 7.315

Reasoning: Write 12.000. Regroup one whole into tenths, then into hundredths and thousandths.

P3 [D-MULT] | student p. 117

Answer: $9.40

Reasoning: 235 cents × 4 = 940 cents, or $9.40.

P4 [D-MULT] | student p. 118

Answer: 0.24; yes

Reasoning: Six tenths of four tenths is twenty-four hundredths.

P5 [D-MULT] | student p. 118

Answer: 0.003

Reasoning: Fifteen hundredths times two hundredths is thirty ten-thousandths, or three thousandths.

P6 [D-MONEY] | student p. 118

Answer: $4.48

Reasoning: Add costs: $5.52. Subtract from $10.00 to get $4.48.

F [D-MULT] | student p. 119

Answer: thousandths; 0.021

Reasoning: Use the worked example in the slides to check each step.

D [D-MULT] | student p. 119

Answer: Six tenths times four tenths is twenty-four hundredths, or 0.24.

Reasoning: He counts digits without tracking the size of the units. Shade six tenths across and four tenths down on a hundred-square grid.

E [D-MULT] | student p. 120

Answer: 0.027

Reasoning: $45 \times 6 = 270$. Hundredths times hundredths makes ten-thousandths: 0.0270. It is less than 0.06.

Watch for 0.27: The product is ten times too large for six hundredths of a number below one.

Long division with decimals

Day 030, student workbook pages 121-126

R1 [R-EQUIV] | student p. 121

Answer: 10 blue tiles

Reasoning: The number of batches is $6 \div 3$. Multiply that count by 5.

Watch for 8 blue tiles: Adding the same amount to both counts changes the ratio.

R2 [R-TABLE] | student p. 121

Answer: 0, 12, and 21 centimeters

Reasoning: Multiply each time by the unit rate. Zero time matches zero distance.

R3 [F-DIV] | student p. 121

Answer: 3

Reasoning: Both fractions use the same unit. Count how many single parts fit.

Watch for $\frac{3}{25}$: Multiplying by the divisor answers a different question.

R4 [R-TABLE] | student p. 121

Answer: 16

Reasoning: Multiply both entries in the first pair by the same factor.

S [EXPLAIN] | student p. 121

Answer: Answers vary. Multiplying the dividend and divisor by the same power of ten preserves the quotient.

Reasoning: Accept an accurate explanation supported by a matching example.

P1 [D-DIV] | student p. 122

Answer: 28

Reasoning: 24 goes into 67 twice, leaving 19 tens. Bring down 2 ones. 192 ÷ 24 = 8.

P2 [D-DIV] | student p. 122

Answer: 3.375

Reasoning: After 3 groups, 3 remains. Rename it as 30 tenths, then continue through hundredths and thousandths.

P3 [D-DIV] | student p. 122

Answer: 1.2

Reasoning: 84 tenths shared among 7 gives 12 tenths.

P4 [D-DIV] | student p. 123

Answer: 70

Reasoning: Multiply both amounts by 100: 630 ÷ 9 = 70.

P5 [D-DIV] | student p. 123

Answer: 0.35

Reasoning: Seven twentieths equals thirty-five hundredths.

P6 [D-DIV] | student p. 123

Answer: 9 pieces

Reasoning: Rename both in tenths: 36 tenths ÷ 4 tenths = 9.

P7 [N-DIV] | student p. 124

Answer: 42

Reasoning: 28 × 40 = 1,120. Two more groups make 56, so 42 groups make 1,176.

F [D-DIV] | student p. 124

Answer: 540; 180

Reasoning: Use the worked example in the slides to check each step.

D [D-DIV] | student p. 124

Answer: Both amounts must scale by 100. Use 630 ÷ 9 = 70.

Reasoning: She changes the group size without changing the available amount by the same factor. Compare 63 tenths with nine hundredths by renaming both in hundredths.

M1 [R-WHOLE] | student p. 125

Answer: $\frac{3}{8}$

Reasoning: Count all eight cups before forming the fraction.

Watch for $\frac{3}{5}$: Five counts the other part, not the whole.

M2 [P-PART] | student p. 125

Answer: 1.8000 dollars

Reasoning: Ten percent is $1.50 and two percent is $0.30. Add them.

M3 [F-DIV] | student p. 125

Answer: $3\ \frac{1}{3}$

Reasoning: Rename the mixed number as an improper fraction, then count groups of three fourths.

Watch for $1\ \frac{7}{8}$: Multiplying by the divisor answers a different question.

E [D-DIV] | student p. 126

Answer: 120

Reasoning: Multiply both by 100: 840 ÷ 7 = 120. Check 120 × 0.07 = 8.4.

Watch for 1.2: Changing only the divisor changes the quotient.

Study guide: Try it

Day 030, student workbook pages 127-130

1 [D-MULT] | student p. 130

Answer: $7.35

Reasoning: 245 cents times three equals 735 cents.

2 [D-MONEY] | student p. 130

Answer: $2.57

Reasoning: The cost is $7.43. Subtract that from $10.00.

3 [D-DIV] | student p. 130

Answer: 1.2

Reasoning: Ninety-six tenths shared among eight gives twelve tenths.

4 [D-DIV] | student p. 130

Answer: 0.225

Reasoning: Use long division into tenths, hundredths, and thousandths.

5 [N-DIV] | student p. 130

Answer: 42

Reasoning: Thirty-two times forty is 1,280. Two more groups add 64.

Daily practice

Day 031, student workbook pages 131-132

1 [N-FACT] | student p. 131

Answer: 21

2 [N-FACT] | student p. 131

Answer: 49

3 [N-FACT] | student p. 131

Answer: 28

4 [N-FACT] | student p. 131

Answer: 63

5 [F-DIV] | student p. 131

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

6 [R-TABLE] | student p. 131

Answer: 32 laps

Reasoning: Four jumps of three minutes match four jumps of eight laps.

7 [N-FACT] | student p. 131

Answer: 24

8 [D-ADD] | student p. 131

Answer: 4.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

9 [D-ADD] | student p. 132 | Built-in review: Decimal practice

Answer: 6.528

Reasoning: Line up decimal points. Add digits in the same place-value columns.

10 [D-MULT] | student p. 132 | Built-in review: Decimal practice

Answer: 0.0050

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0500: The decimal point is one place too far right. Check the product against both factors.

11 [D-SUB] | student p. 132 | Built-in review: Decimal practice

Answer: 7.714

Reasoning: Write 10.000 and regroup through the zero columns.

12 [D-MULT] | student p. 132 | Built-in review: Decimal practice

Answer: 0.56; yes

Reasoning: Eight tenths of seven tenths covers fifty-six hundredths.

13 [D-MULT] | student p. 132

Answer: Six tenths times four tenths is twenty-four hundredths, or 0.24.

Watch for 2.4: He counts digits without tracking the size of the units.

Daily practice

Day 032, student workbook pages 133-134

1 [F-EQUIV] | student p. 133

Answer: 0.5; 50%

2 [F-EQUIV] | student p. 133

Answer: 0.25; 25%

3 [F-EQUIV] | student p. 133

Answer: 0.75; 75%

4 [F-EQUIV] | student p. 133

Answer: 0.125; 12.5%

5 [P-WHOLE] | student p. 133

Answer: 60 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 5.40 shells: Taking 30% of the known part does not recover the whole.

6 [F-SUB] | student p. 133

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

7 [N-FACT] | student p. 133

Answer: 12

8 [D-DIV] | student p. 133

Answer: 20

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.2: Changing only the divisor changes the quotient. Scale both quantities by 100.

9 [N-DIV] | student p. 134 | Built-in review: Multiplication and long division

Answer: 13

Reasoning: Use long division and verify quotient × 24 equals the dividend.

10 [N-MULT] | student p. 134 | Built-in review: Multiplication and long division

Answer: 1404

11 [N-DIV] | student p. 134 | Built-in review: Multiplication and long division

Answer: 42

Reasoning: Thirty-two times forty is 1,280. Two more groups add 64.

12 [N-DIV] | student p. 134 | Built-in review: Multiplication and long division

Answer: 13

13 [D-DIV] | student p. 134

Answer: Both amounts must scale by 100. Use 630 ÷ 9 = 70.

Watch for 0.7: She changes the group size without changing the available amount by the same factor.

Daily practice

Day 033, student workbook pages 135-136

1 [N-SCALE] | student p. 135

Answer: 72

2 [N-SCALE] | student p. 135

Answer: 7

3 [N-SCALE] | student p. 135

Answer: 15

4 [N-SCALE] | student p. 135

Answer: 21

5 [D-DIV] | student p. 135

Answer: 30

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.3: Changing only the divisor changes the quotient. Scale both quantities by 100.

6 [D-MULT] | student p. 135

Answer: 0.0075

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0750: The decimal point is one place too far right. Check the product against both factors.

7 [P-PART] | student p. 135

Answer: 1.8000 dollars

Reasoning: Ten percent is $1.50 and two percent is $0.30. Add them.

8 [P-WHOLE] | student p. 135

Answer: 80 shells

Reasoning: Divide the given amount into three 10% groups, then take ten such groups.

Watch for 7.20 shells: Taking 30% of the known part does not recover the whole.

9 [F-SUB] | student p. 136

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

10 [R-EQUIV] | student p. 136

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

11 [N-FACT] | student p. 136

Answer: 15

12 [G-VOL] | student p. 136

Answer: 45 cm³

13 [PUZZLE] | student p. 136 | Built-in review: Weekly puzzle

Answer: 4.82

Reasoning: A ones digit of 4 places the number nearest 5. Among 4.28 and 4.82, the larger is nearer to 5.

Daily practice

Day 034, student workbook pages 137-138

1 [N-EST] | student p. 137

Answer: 700

2 [N-EST] | student p. 137

Answer: 8

3 [N-EST] | student p. 137

Answer: About 100

4 [N-EST] | student p. 137

Answer: About 20

5 [D-DIV] | student p. 137

Answer: 30

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.3: Changing only the divisor changes the quotient. Scale both quantities by 100.

6 [D-MULT] | student p. 137

Answer: 0.0075

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0750: The decimal point is one place too far right. Check the product against both factors.

7 [D-DIV] | student p. 137

Answer: 0.875

Reasoning: Continue the long division into tenths, hundredths, and thousandths until the remainder is zero.

8 [F-DIV] | student p. 137

Answer: 2

Reasoning: Both fractions use the same unit. Count how many single parts fit.

Watch for $\frac{1}{8}$: Multiplying by the divisor answers a different question.

9 [F-DIV] | student p. 138 | Built-in review: Fraction operations

Answer: 20; $\frac{1}{20}$

Reasoning: The first counts fifths in four. The second splits one fifth into four shares.

Watch for $\frac{4}{5}$; $\frac{4}{5}$: Changing the order changes what is shared or counted.

10 [F-MULT] | student p. 138 | Built-in review: Fraction operations

Answer: $2\ \frac{2}{3}$

Reasoning: Count two thirds for each whole-number group, then rename the result.

11 [F-MULT] | student p. 138 | Built-in review: Fraction operations

Answer: $\frac{4}{15}$

12 [F-MULT] | student p. 138 | Built-in review: Fraction operations

Answer: $\frac{4}{5}$

13 [D-MONEY] | student p. 138

Answer: $2.57

Reasoning: The cost is $7.43. Subtract that from $10.00.

14 [D-DIV] | student p. 138

Answer: Both amounts must scale by 100. Use 630 ÷ 9 = 70.

Watch for 0.7: She changes the group size without changing the available amount by the same factor.

Weekly check

Day 035, student workbook pages 139-140

1 [D-DIV] | student p. 139

Answer: 30

Reasoning: Multiply both dividend and divisor by 100, then divide by 7.

Watch for 0.3: Changing only the divisor changes the quotient. Scale both quantities by 100.

2 [D-MULT] | student p. 139

Answer: 0.0075

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.0750: The decimal point is one place too far right. Check the product against both factors.

3 [D-DIV] | student p. 139

Answer: 0.875

Reasoning: Continue the long division into tenths, hundredths, and thousandths until the remainder is zero.

4 [D-MULT] | student p. 139

Answer: 0.0100

Reasoning: One fourth of the second factor must be smaller than it. Use this to check the decimal position.

Watch for 0.1000: The decimal point is one place too far right. Check the product against both factors.

5 [F-DIV] | student p. 139

Answer: $\frac{5}{24}$ liter

Reasoning: Split each sixth into four equal parts. Each cup gets five twenty-fourths.

6 [R-RATE] | student p. 140

Answer: 4 labels per minute; $\frac{1}{4}$ minute per label

Reasoning: Divide labels by minutes for the first rate. Reverse for the second.

7 [F-DIV] | student p. 140

Answer: $\frac{3}{4}$

Reasoning: One half is three sixths. A group is four sixths. Three of four group parts fit.

Watch for $\frac{1}{3}$: The proposed answer multiplies instead of dividing.

8 [F-SUB] | student p. 140

Answer: $\frac{5}{12}$

Reasoning: Rename both fractions with a common denominator. Subtract equal parts.

9 [R-EQUIV] | student p. 140

Answer: 4 : 7

Reasoning: Divide both counts by the same factor.

10 [D-MULT] | student p. 140

Answer: Six tenths times four tenths is twenty-four hundredths, or 0.24.

Watch for 2.4: He counts digits without tracking the size of the units.